

Machine Learning for Survival Prediction in Breast Cancer using TCGA gene expression data

Tungalan Ganbaatar

BS845

Link to R work



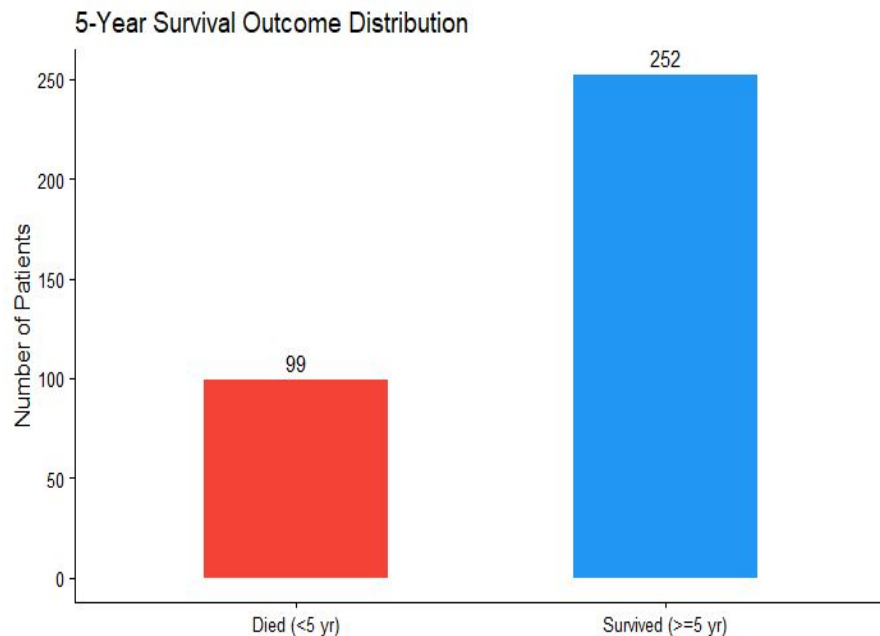
Link to PPT



Goal & Data

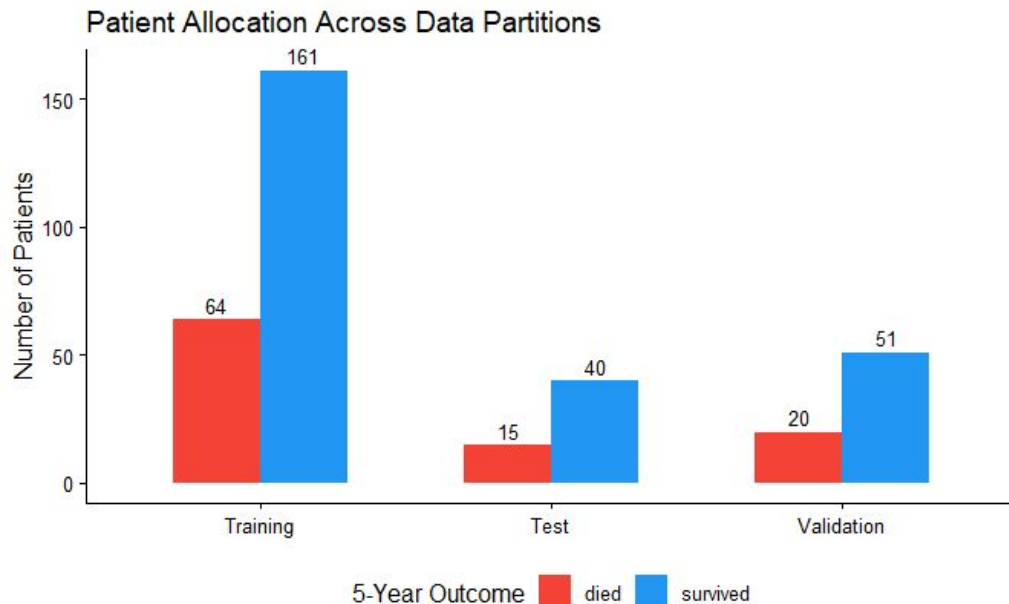
Can gene expression predict 5-year survival in breast cancer?

- TCGA-BRCA dataset
- 351 patients with gene expression and clinical data
- Outcome: 0 = died (<5 yr), 1 = survived (≥ 5 yr)
- Imbalanced dataset (~72% survived)



Preprocessing

- Log-transform gene expression
- Select top 500 genes (variance) to reduce high dimensionality ($p \gg n$)
- 3-way split (train/test/validation)



Final dataset: 351 patients × 500 genes

Modeling Approach

Ridge (logistic + L2)

→ handles high-dimensional data

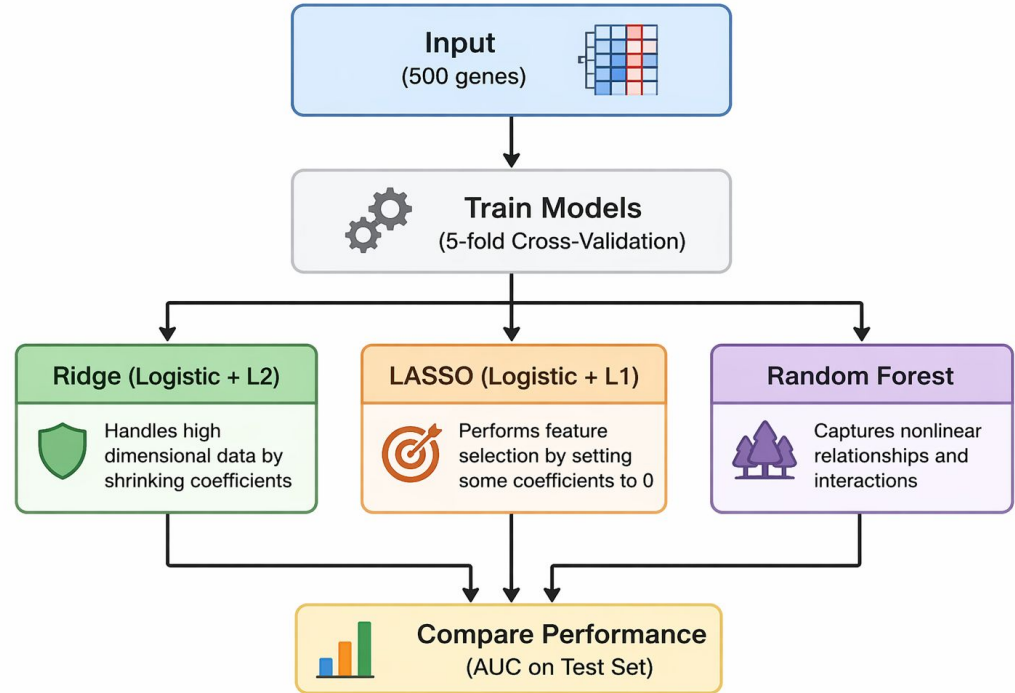
LASSO (logistic + L1)

→ feature selection

Random Forest

→ captures nonlinear relationships and interactions

5-fold cross-validation used for tuning



Results (Model Performance)

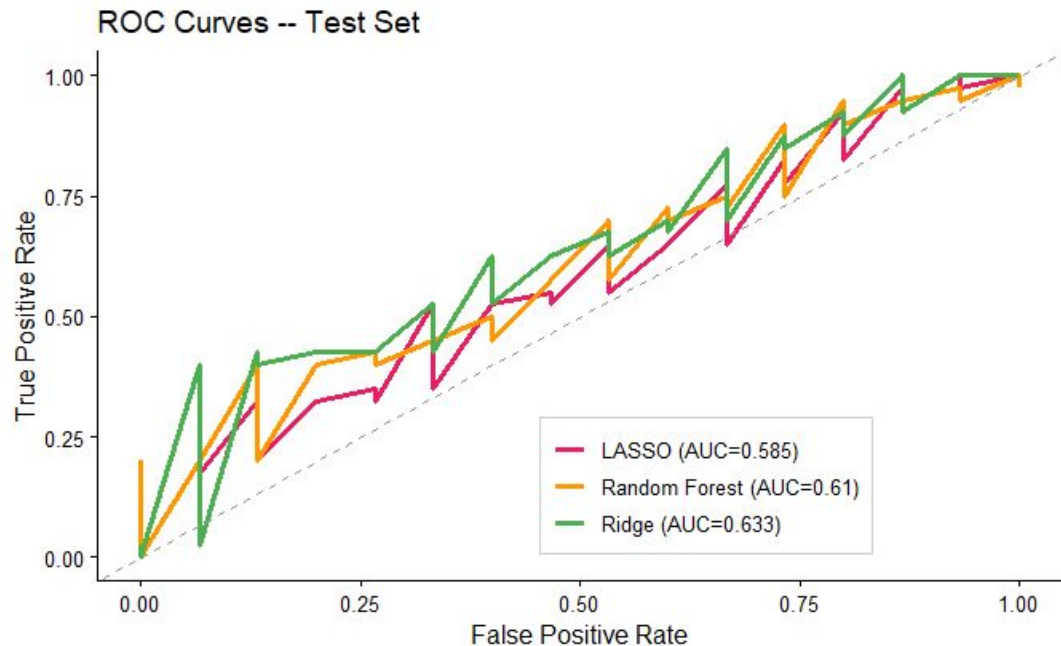
Ridge: AUC = **0.633 (test)**, **0.727 (validation)**

Random Forest: AUC = 0.610

LASSO: AUC = 0.585

Ridge performs best, overall predictive power is modest!

AUC used due to class imbalance

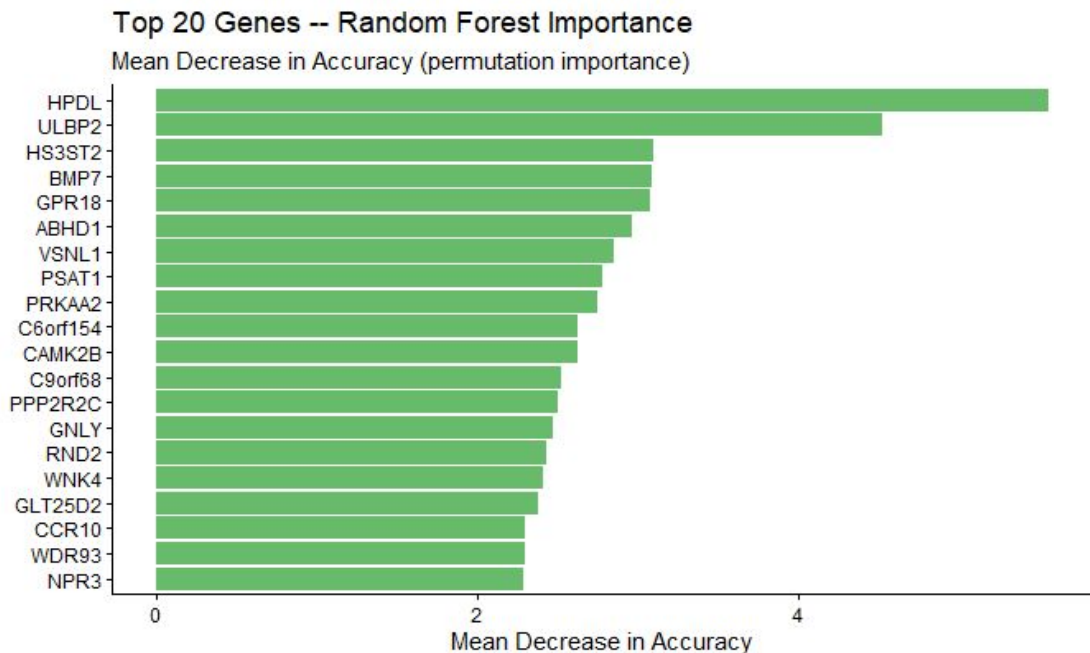


Interpretation & Insights

LASSO selected 70 key genes → feature selection

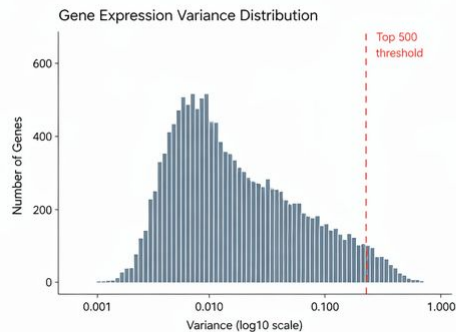
- Random Forest identifies important genes (nonlinear effects)
- 2 genes overlap across methods (**GLT25D2**, **BMP7**) → more robust signals
- Models tend to predict survival due to class imbalance

Different models capture different aspects of the data!

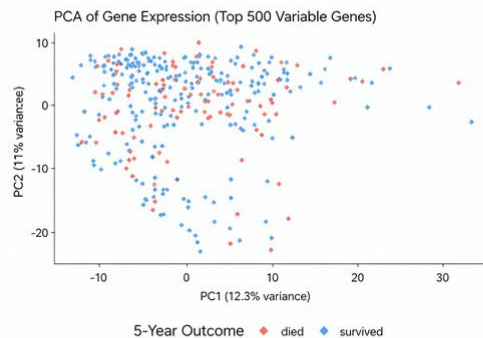


Conclusion

- Ridge achieved the best performance (AUC $\approx$ 0.63 test, 0.73 validation)
- Overall predictive power is **modest**
- Gene expression contains signal, but is **high-dimensional and complex**
- Feature selection and model comparison provided biological insights
- Future work: larger datasets or more advanced models



Thank You!



I'm happy to take
any questions
you have!

